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Chapter 1

Methods — Forecasting Models and Evaluation

1.1 Replication framing

- One paragraph: methodology replicated from [3] (LEAR, DNN) and [4] (Transformer) on current market data; absolute errors are not comparable across periods, so comparison uses rMAE and model rankings ([3] §5.4).
- State the three models and two markets; all six runs share one test window so DM/GW tests are paired.

1.2 Data and features

Table 1.1: Datasets. Both markets span 2022-03-10 to 2026-05-31 (ENTSO-E).

Market	**Resolution**	**Exogenous (paper-exact)**	**Paper analogue**
DK1	hourly (24)[a]	TotalLoad + Wind (on+offshore)	Nord Pool: load + wind [3]
DE	hourly (24)[a]	TotalLoad + Renewables (wind+solar)	EPEX-DE: load + wind&solar [3]

[a] Both markets clear in 15-min blocks (DE throughout; DK1 since 2025-10-01); prices and exogenous forecasts are downsampled t

- Feature set per [3] Eq. 3: price lags $d{-}1, d{-}2, d{-}3, d{-}7$; exogenous at $d0, d{-}1, d{-}7$; 7 day-of-week dummies ($4n + 2 \cdot 3n + 7$; n=24 ⇒ 247, n=96 ⇒ 967).
- Aggregation: raw ENTSO-E wind/solar curves summed at load time (`exogenous_aggregates`); members inner-joined on timestamps before summing; all exogenous columns inner-joined so the coarsest series anchors the time grid.

- Variance stabilisation: asinh transform [5].
- Resolution: both markets clear in 15-minute blocks (DE throughout the sample; DK1 since 2025-10-01). To keep the two markets comparable with each other and with the hourly reference benchmarks [3, 4], prices and exogenous forecasts are downsampled to an arithmetic hourly mean (24 periods/day) before feature construction. Modelling natively at 96 periods is infeasible for DK1, which has under one month of 15-min history before the test window.
- Data-gap note: ENTSO-E lacks DK1 wind forecasts for a handful of days (incl. 2026-05-07 inside the test window); affected days drop from all DK1 models identically, so paired tests remain valid.

1.3 Models

Table 1.2: Model setups (one row per model; identical across both markets unless noted).

Model	Spec	Calibration	Refit
LEAR	One Lasso per delivery period; λ by `LassoCV` (time-series CV)	ensemble over 56/84/1092 d	daily
DNN	MLP, 4-seed ensemble; architecture HPs by Optuna TPE (50 trials, ex-ante val)	730 d	daily
Transformer	Encoder stack per [4]; paper NP hyperparameters	1092 d	weekly

- Tuning protocol: validation window 2025-01-01 to 2025-10-23 (42 weeks, immediately preceding the test window; [3] §4.3.2); trials persisted to `algo.hyperparam_trials`.
- Transformer calibration windows verified against usable sequence rows before test start (DK1: 1197, DE: 1265 — both $\geq$ 1092).

1.4 Evaluation protocol

- Test window 2025-10-24 to 2026-05-31 (220 calendar days), identical for all six runs; training data = everything before test start ([3] §5.7).
- Metrics: MAE, RMSE, sMAPE, rMAE with weekly-naive ([3] §5.4); rMAE is the headline metric.
- Statistical testing: multivariate GW on daily L^1 loss differentials with Newey–West HAC errors [2, 3]; pairwise DM [1] as univariate complement. Models compared on their common surviving test days (feature pipelines drop gap/DST days differently per model family).
- Computation cost: training seconds per recalibration day, from `algo.experiment_runs` (analogue of [3] Table 4).

1.5 Deviations from the source papers

Table 1.3: Documented deviations from [3] and [4].

Aspect	**Paper**	**This work (reason)**
Test length	$\geq$ 1–2 years	31 weeks (price history starts 2022-03; val + calibration consume the rest)
DNN tuning budget	1500 TPE iterations	50 trials (compute; optimum stable early in prior studies)
DNN search space	architecture + feature selectors	architecture only (feature set fixed to paper spec)
LEAR windows	56/84/1092/1456 d	56/84/1092 d (1456 predates data)
DNN window	4 years	730 d (data/compute constraint)
λ selection	LARS + AIC	`LassoCV` time-series CV (standard, equivalent accuracy)
Market resolution	hourly (paper era)	both markets now 15-min; downsampled to an hourly mean for cross-market and cross-paper comparability
Transformer pipeline	train-once + renormalise	calibration window + weekly refit (pipeline consistency)
Transformer HPs	tuned per dataset	paper NP optimum reused (future work)

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